

Skin Cancer Lesion Segmentation with U-Net + ITTA-TLBO Optimization

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Abstract

This project applies deep learning and hybrid metaheuristic optimization to the problem of skin cancer lesion segmentation. We combine a U-Net convolutional neural network with a threshold optimization step powered by a hybrid Improved Tiki-Taka Algorithm (ITTA) and Teaching-Learning Based Optimization (TLBO). Experimental evaluation on the Skin Cancer Lesions Segmentation dataset demonstrates that the proposed framework improves segmentation accuracy by adapting the probability threshold for binary mask generation.

Keywords: U-Net, ITTA, TLBO, Skin Cancer, Segmentation, Dice Score, Deep Learning, Optimization.

1 Introduction

Skin cancer is one of the most prevalent cancers worldwide, and early detection plays a critical role in improving survival rates. Dermoscopic imaging has become a standard tool for diagnosing skin lesions, but accurate segmentation of lesions remains challenging due to variations in shape, size, color, and low contrast with surrounding skin. Convolutional Neural Networks (CNNs), particularly U-Net architectures, have proven effective in medical image segmentation. However, fixed thresholding of predicted probability maps can limit performance. In this work, we propose to optimize the threshold using a hybrid Improved Tiki-Taka Algorithm with Teaching-Learning Based Optimization (ITTA-TLBO).

2 Dataset

- **Source:** Skin Cancer Lesions Segmentation (Kaggle)
- **Size:** 10,015 images with ground truth masks
- **License:** CC-BY-NC-SA-4.0

3 Methodology

3.1 Data Preparation

- Images and masks resized to 128x128.
- Pixel values normalized.
- Dataset split into 80% training and 20% validation.

3.2 Model Architecture: U-Net

- Encoder-decoder CNN with skip connections.
- Binary cross-entropy loss with Adam optimizer.
- Trained for 10 epochs, batch size 16.

3.3 Optimization

Instead of fixing the threshold at 0.5, ITTA-TLBO is applied to optimize the threshold value for binarizing U-Net outputs. The optimization objective is to maximize the Dice Similarity Coefficient (DSC) between predicted and ground truth masks.

4 Results

- Best Dice Score: **0.888**
- Optimal Threshold: **0.4611**
- Validation Accuracy: **93–94%**

5 Conclusion

We presented a framework that combines U-Net segmentation with ITTA-TLBO threshold optimization for skin cancer lesion detection. The optimized threshold significantly improves segmentation performance compared to a fixed 0.5 threshold. This approach highlights the potential of hybrid deep learning and optimization methods in medical image analysis.